

Ejer localizacion ideal primo anillo local

Ejercicio 1. Sea R un anillo y sea $\mathfrak{p} \subset R$ un ideal primo. Demuestra que $R_{\mathfrak{p}}$ es un anillo local con ideal maximal $\mathfrak{p}R_{\mathfrak{p}}$.

Solución: Queremos probar que $R_{\mathfrak{p}}$ tiene un único ideal maximal que es $\mathfrak{p}R_{\mathfrak{p}}$. Sea $\mathfrak{m} \subset R_{\mathfrak{p}}$ un ideal maximal. Entonces, en particular $\mathfrak{m}$ es un ideal primo, luego por el `cor-ideal-primo-localizacion-extendido/Corolario 1`, $\exists \mathfrak{q} \subset R$ ideal primo tal que $\mathfrak{m} = \mathfrak{q}^e$ y $\mathfrak{q} \cap S = \emptyset$. Pero como $S = R \setminus \mathfrak{p}$, tenemos que $\mathfrak{q} \subset \mathfrak{p}$.

Como $\mathfrak{q} \subset \mathfrak{p}$, tenemos que $\mathfrak{q}^e \subset \mathfrak{p}^e$, es decir, $\mathfrak{m} \subset \mathfrak{p}^e$. Además, como $\mathfrak{p} \cap S = \emptyset$, tenemos que $\mathfrak{p}^e \neq R_{\mathfrak{p}}$. Por tanto, $\mathfrak{m} \subset \mathfrak{p}^e \subsetneq R_{\mathfrak{p}}$. Como $\mathfrak{m}$ es maximal, debe ser $\mathfrak{m} = \mathfrak{p}^e = \mathfrak{p}R_{\mathfrak{p}}$. ■

Referenciado en

- `teo-extension-entera-ideal-primo-imp-exists-ideal-primo-contrae`

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